Shape-First Behavioral Segmentation of Household Energy Consumption: A Reproducible Framework with Evidence-Based K-Selection and Controlled Validation: A Reproducible Synthetic Benchmark

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Abstract

Household energy consumption segmentation typically groups consumers by total usage magnitude, overlooking temporal behavioral patterns critical for demand response and grid management. We present a reproducible framework that separates consumption magnitude from temporal usage behavior through shape-normalized feature engineering, enabling behavioral clustering that reflects when energy is consumed rather than how much. The framework employs 51 scale-invariant features derived from normalized 24-hour load profiles, applies PCA to retain 95% variance (10 components), and selects the number of clusters K via a pre-specified multi-criterion composite rule combining silhouette coefficient, Calinski-Harabasz index, Davies-Bouldin index, and multi-seed stability analysis. Validated on a controlled synthetic dataset of 200 consumers over 365 days with known ground-truth behavioral archetypes, the framework achieves K=4 with Adjusted Rand Index (ARI) 0.813 against hidden labels never used during clustering. Multi-seed stability yields mean pairwise ARI 0.995, and temporal stability across quarterly segments achieves mean ARI 0.882, demonstrating robust behavioral groupings independent of seasonal magnitude variations. A comprehensive ablation study shows magnitude-only features achieve near-zero archetype recovery (ARI approximately 0.00) despite highest silhouette scores (0.52), confirming successful separation of scale from behavior. The framework includes complete reproducibility infrastructure with cryptographic configuration hashing, deterministic execution, versioned artifacts, and documented pathways

for real-world data adaptation. All code, synthetic data generation, experimental protocols, and results are publicly available. This work contributes methodological rigor and reproducibility standards for behavioral energy analytics rather than novel clustering algorithms.

Keywords: household energy consumption, load profile clustering, behavioral segmentation, shape normalization, PCA, K-means, evidence-based K selection, reproducible research, smart meter analytics, demand response

1 Introduction

The widespread deployment of smart meters has enabled high-resolution monitoring of household electricity consumption at unprecedented scale [1, 2]. This granular data enables segmentation of consumers into behaviorally distinct groups, supporting targeted demand response programs, dynamic pricing strategies, and infrastructure planning [3, 4]. However, traditional clustering approaches often group households primarily by total consumption magnitude—how much energy they use—rather than by temporal usage patterns—when energy is consumed [5, 6].

Magnitude-based segmentation conflates two independent dimensions: a household's energy intensity (driven by building size, occupancy, appliances) and its behavioral routine (driven by occupancy patterns, work schedules, lifestyle preferences). Two households with identical daily load shapes but different consumption scales will be separated by magnitude-driven clustering, while households with opposite peak hours but similar totals may be grouped together. For demand response applications, behavioral timing is often more actionable than aggregate consumption [7].

1.1 Motivation

Consider two households: one peaks at midday (home office, daytime HVAC), another peaks at evening (commuter, evening cooking and entertainment). If the first consumes 30 kWh/day and the second 25 kWh/day, magnitude-driven clustering may separate them despite nearly identical timing. Conversely, if both consume 30 kWh/day, they may cluster together despite opposite demand profiles. For a utility designing time-of-use rates or load-shifting incentives, the temporal pattern is the actionable signal [8, 9].

Separating magnitude from timing requires explicit normalization: transforming each household's 24-hour load curve into a shape representation invariant to scalar multiplication. While shape-based clustering has been explored [10, 5], comprehensive frameworks addressing feature engineering, dimensionality reduction, evidence-based K selection, stability validation,

and reproducibility remain limited in the literature.

1.2 Research Gap

Existing household energy clustering studies face several methodological challenges:

1. **Feature Engineering:** Many approaches use raw hourly values or magnitude-contaminated features, allowing consumption scale to dominate distance calculations [2, 6].
2. **K Selection:** The number of clusters K is often chosen heuristically (e.g., 3-5 for interpretability) or via single metrics (elbow method, silhouette alone) without systematic evaluation [11].
3. **Validation:** Internal metrics (silhouette, Calinski-Harabasz) measure cluster quality but cannot confirm meaningful behavioral distinctions. External validation requires ground truth rarely available in real datasets [12].
4. **Stability:** Cluster robustness across random initializations and temporal windows is often unreported, leaving uncertainty about partition reliability [13].
5. **Reproducibility:** Many studies lack publicly available code, data, or sufficient methodological detail for independent replication [14].

1.3 Contributions

This paper addresses these gaps through a comprehensive, reproducible framework for shape-first behavioral segmentation. Our specific contributions are:

1. **Shape-Normalized Feature Engineering:** 51 scale-invariant behavioral features derived from normalized 24-hour load profiles, explicitly separating magnitude from temporal patterns.
2. **Evidence-Based K Selection:** Pre-specified multi-criterion composite rule combining silhouette, Calinski-Harabasz, Davies-Bouldin, cluster balance constraints, and multi-seed stability, with parsimony preference when solutions are statistically indistinguishable.
3. **Controlled Validation:** Synthetic dataset with known ground-truth behavioral archetypes enabling quantitative assessment of recovery (ARI=0.813), while explicitly hiding ground truth during all modeling steps.

4. **Comprehensive Stability Analysis:** Multi-seed clustering stability (mean ARI=0.995) and temporal stability across quarterly segments (mean ARI=0.882), confirming robust behavioral groupings independent of seasonal variations.

5. **Rigorous Ablation Study:** Systematic comparison of five feature sets (magnitude-only, shape-only, behavioral, combined) demonstrating that magnitude-only features achieve near-zero behavioral recovery (ARI approximately 0.00) despite highest internal quality metrics.

6. **Complete Reproducibility Infrastructure:** Deterministic configuration with cryptographic hashing, pinned dependencies, versioned artifacts, documented execution protocols, and publicly available implementation.

1.4 Scope and Positioning

This work is a methodology paper emphasizing reproducible research practices for behavioral energy analytics. We do not claim novelty in PCA or K-means algorithms (both well-established [15, 16, 17]), but rather contribute rigorous implementation, comprehensive validation, and reproducibility standards. The primary experimental validation uses synthetic data with known ground truth; a documented pathway for real-world data exists but is not executed in this study (discussed in Section 7).

1.5 Paper Organization

Section 2 reviews related work on household energy clustering, PCA-based dimensionality reduction, and validation methodologies. Section 3 details the synthetic dataset generation and provenance. Section 3 presents the complete analytical pipeline from feature engineering through cluster profiling. Section 4 describes experimental design and evaluation protocols. Section 5 presents flagship results, ablation studies, and stability analyses. Section 6 interprets findings and practical implications. Section 7 acknowledges methodological constraints and dataset scope. Section 8 summarizes contributions and future directions.

2 Literature Survey

2.1 Household and Building Energy Consumption Analysis

Smart meter deployment has enabled fine-grained analysis of residential electricity consumption [2, 6]. Early work focused on aggregated load profiles for utility-level forecasting [7], but increasing attention has shifted to disaggregated household-level patterns revealing behavioral heterogeneity [1, 5]. Kwac et al. [2] applied hierarchical clustering to hourly smart meter data, identifying distinct consumption patterns but noting that magnitude variations dominated the groupings. Rhodes et al. [6] used K-means on raw consumption features, achieving interpretable clusters but without systematic separation of scale from shape. Jin et al. [5] at Lawrence Berkeley National Laboratory developed foundational clustering methodologies for residential smart meter data, emphasizing the importance of feature engineering and validation.

Recent work increasingly recognizes the value of shape-based clustering. Lazar et al. [1] applied adaptive K-means to daily load shapes, finding outdoor temperature as the dominant driver of variability in summer-peaking utilities. Afzalan et al. [8] proposed a two-stage approach separating temporal patterns from peak demand using Dynamic Time Warping, explicitly addressing shape misalignment. Valdes et al. [9] introduced two-stage k-medoids clustering of normalized load shapes for demand response in high-renewable contexts.

2.2 Load Profile and Electricity Consumption Clustering

Clustering methodologies for electricity consumption profiles span multiple approaches. Chicco et al. [18] integrated socio-demographic survey data with smart meter clustering using K-means. Favre-Bulle [19] emphasized high-frequency data for heterogeneity analysis in residential load profiling. These studies demonstrate that clustering can reveal meaningful consumer segments, but most rely on magnitude-contaminated features or lack systematic validation of behavioral recovery.

2.3 Behavioural and Shape-Based Load Profile Analysis

Several studies explicitly address magnitude-timing separation. Rahayu et al. [10] proposed shape-based clustering using Dynamic Time Warping to align temporal patterns independent of magnitude. Afzalan et al. [8] extracted centroids accounting for shape misalignments before merging clusters of similar magnitude. Valdes et al. [9] normalized daily time series to

unit sum, enabling clustering by temporal distribution rather than absolute consumption.

However, most approaches focus on shape alignment or normalization of raw hourly sequences, without systematic engineering of scale-invariant behavioral descriptors (e.g., peak concentration, harmonic structure, weekend behavior, load variability). Our work contributes a comprehensive 51-feature behavioral representation explicitly designed for magnitude-timing separation.

2.4 Dimensionality Reduction and PCA in Energy Analytics

Principal Component Analysis (PCA) is widely used for dimensionality reduction in energy analytics [15]. Haben et al. [20] applied functional PCA to electric consumption patterns for multi-horizon forecasting, demonstrating PCA's effectiveness in capturing temporal load structure. Liao et al. [21] combined PCA with logistic regression for pattern-based energy consumption analysis, using PCA to mitigate multicollinearity among correlated features.

Ding and He [22] established theoretical connections between K-means and PCA, showing that K-means cluster indicators span the principal subspace under certain conditions. This relationship motivates PCA as a preprocessing step for K-means, reducing noise and computational cost while preserving cluster-relevant variance.

2.5 K-Means and Alternative Clustering Approaches

K-means clustering [16] with k-means++ initialization [17] remains the most widely used method for energy consumption segmentation due to its simplicity and scalability. However, K-means assumes spherical clusters of equal size, which may not hold for real consumption patterns. Alternative methods include hierarchical clustering [6], Gaussian mixture models, and density-based approaches, though these are less commonly applied in energy literature due to computational complexity or interpretability challenges.

2.6 Cluster Validation and Stability

Determining optimal K remains a fundamental challenge in unsupervised clustering [11]. Common heuristics include elbow methods (inertia curve inflection), silhouette analysis [23], and variance-ratio criteria (Calinski-Harabasz [24]). Davies-Bouldin index [25] measures cluster compactness versus separation. However, single-metric approaches may yield conflicting recommendations, and internal metrics cannot confirm semantic meaningfulness [12].

Hennig and Liao [11] proposed aggregating multiple calibrated validity indices, recognizing that different metrics capture complementary aspects of cluster quality. Gates and Ahn [12] reviewed partition comparison indices including Adjusted Rand Index (ARI), emphasizing permutation-invariance and correction for chance agreement. Henning et al. [13] proposed Modified ARI (MARI) based on multinomial rather than hypergeometric models, addressing bias in dependent clusterings.

Clustering stability across random initializations is critical for reliable findings [13]. Pairwise ARI between partitions from different seeds quantifies reproducibility independent of ground truth. Temporal stability—assessing whether cluster structure persists across time windows—has received less attention in energy literature despite its practical importance for longitudinal demand response.

2.7 Research Gap and Motivation

Despite substantial prior work, several gaps remain:

- Most studies use magnitude-contaminated features, conflating scale with behavior
- K selection is often heuristic rather than evidence-based
- Internal validation metrics cannot confirm behavioral meaningfulness
- Stability across seeds and time windows is rarely quantified
- Reproducibility infrastructure (code, data, protocols) is limited

Table 1 summarizes key prior studies and their limitations relative to the current work.

Our framework addresses these gaps through systematic magnitude-timing separation, pre-specified multi-criterion K selection, controlled validation with synthetic ground truth, comprehensive stability analysis, and complete reproducibility infrastructure.

3 Dataset and Methodology

3.1 Source Dataset and Data Provenance

This study uses a fully synthetic dataset generated within the repository. There is no original real-world dataset serving as a basis. The synthetic data

Table 1: Comparison of Prior Household Energy Clustering Studies

Study	Data/Representation	Clustering Approach	Validation	Limitation/Relevance
Kwac et al. (2014) [2]	Hourly smart meter	Hierarchical clustering	Internal metrics only	Magnitude dominates groupings
Rhodes et al. (2014) [6]	Raw consumption features	K-means	Internal metrics only	No magnitude-timing separation
Jin et al. (2016) [5]	Smart meter data	Multiple methods	Internal metrics only	Feature engineering focus, no stability
Lazar et al. (2021) [1]	Daily load shapes	Adaptive K-means	Temperature correlation	No temporal stability validation
Afzalan et al. (2020) [8]	Load shapes	Two-stage DTW + K-means	Internal metrics only	No controlled ground truth
Valdes et al. (2021) [9]	Normalized shapes	K-medoids	Internal metrics only	No multi-criterion K selection
This study	**51 behavioral features**	**K-means + PCA**	**Synthetic ground truth + stability**	**Magnitude-timing separation, reproducibility**

generator is implemented in `src/data_loader.py` and is designed to produce controlled, reproducible household energy consumption patterns with known ground-truth behavioral archetypes.

Rationale for Synthetic Data: Real household smart meter data with known behavioral ground truth is unavailable due to privacy constraints and absence of behavioral labels. Synthetic data enables controlled validation: each consumer is assigned a behavioral archetype, allowing quantitative assessment of clustering recovery while maintaining the independence property critical for evaluating magnitude-timing separation.

Generator Design Principles: The synthetic generator is hand-designed to create four behavioral archetypes with realistic temporal patterns. Two critical design choices ensure the experiment tests magnitude-timing separation:

1. **Magnitude Independence:** Amplitude (how much a consumer uses) is drawn from the same log-normal distribution for every archetype. Magnitude therefore carries no archetype information, enabling the ablation study to demonstrate that magnitude features cannot recover the groups even when they score well on internal metrics.

2. **Archetype Overlap:** Consumers are blended towards the population-average shape by a random amount drawn from a Beta(1.4, 5.0) dis-

tribution (mean approximately 0.22). This creates realistic overlap between archetypes, making the clustering problem non-trivial.

3.2 Synthetic Dataset Generation

The synthetic generator produces hourly electricity consumption for N consumers over D days. Each consumer is assigned to one of four behavioral archetypes: daytime, evening, flat, or weekend.

3.2.1 Archetype Definitions

The four archetypes are defined by 24-hour load shape templates:

- **Daytime:** Peaks 12:00-14:00, lower evening demand (home office, midday HVAC)
- **Evening:** Peaks 18:00-21:00, lower daytime demand (commuter, evening cooking/entertainment)
- **Flat:** Approximately uniform 24-hour distribution (commercial-like, always-on baseline)
- **Weekend:** Evening-peaking with elevated weekend consumption (lifestyle-driven weekly variation)

Each archetype is specified by Gaussian bump parameters (amplitude, center hour, width) for morning, midday, and evening peaks, plus weekend energy ratio and weekend shift hours. The full specifications are defined in `ARCHETYPE_SPECS` in `src/data_loader.py`.

3.2.2 Generative Model

For consumer i at hour h on day d, the consumption $E_{i,h,d}$ (in kWh) is modeled as:

$$E_{i,h,d} = B_i \cdot S_{\mathcal{A}_i}(h, d) \cdot M_i(d) \cdot (1 + \epsilon_{i,h,d}) \tag{1}$$

where:

- $B_i \sim \text{LogNormal}(\log(1.2), 0.45)$ is base consumption magnitude (kW), drawn independently of archetype $\mathcal{A}_i$

- $S_{\mathcal{A}_i}(h, d)$ is the normalized 24-hour shape template for archetype $\mathcal{A}_i$ (sums to 1)
- $M_i(d)$ is the seasonal magnitude multiplier for consumer i on day d (mean-corrected, independent of $\mathcal{A}_i$)
- $\epsilon_{i,h,d} \sim \mathcal{N}(0, \sigma^2)$ is Gaussian noise with consumer-specific σ_i

3.2.3 Seasonal Variation

The generator includes an optional seasonal model (enabled by default for the flagship run) with two separable channels:

- **Magnitude channel:** Daily total demand swings by ±25% around the consumer's long-run mean. The factor is mean-corrected over the observation window, so it changes when energy is used, not how much on average.
- **Shape/timing channel:** On the consumer's peak day, the morning peak moves 1 hour earlier and the evening peak moves 1 hour later (long days spread the daily rhythm out); half a year later the pattern is reversed.

Both channels share one per-consumer phase drawn from a normal distribution centered on the northern hemisphere summer solstice (day 172) with standard deviation 20 days. Phase is drawn independently of archetype, so timing shifts do not create spurious clusters.

3.2.4 Dataset Specifications

The flagship experiment uses the following configuration:

- Number of consumers: 200
- Observation period: 365 days (2024-01-01 to 2024-12-30)
- Sampling interval: Hourly (8,760 hours per consumer)
- Total records: 1,752,000 hourly measurements
- Archetype distribution: Equal (50 consumers per archetype)
- Random seed: 42
- Configuration hash: 99c7a6631340d301

3.2.5 Ground Truth Handling

Archetype labels and seasonal phases are stored in the generated dataset but dropped before any preprocessing or modeling. These labels are used exclusively for post-hoc validation via Adjusted Rand Index (ARI) and never influence feature engineering, PCA, or K-means. This independence is verified through dataset integrity checks.

3.3 Feature Engineering

Feature engineering is the core contribution, separating magnitude from temporal behavior through explicit normalization.

3.3.1 Shape Normalization

For each consumer i, compute mean hourly consumption $\mu_{i,h} = \frac{1}{D}\sum_{d=1}^{D} E_{i,h,d}$ across all days. The normalized 24-hour shape is:

$$\tilde{S}_{i,h} = \frac{\mu_{i,h}}{\sum_{h'=0}^{23} \mu_{i,h'}} \tag{2}$$

This transformation ensures $\sum_{h=0}^{23} \tilde{S}_{i,h} = 1$ and makes the shape invariant to scalar multiplication: $\alpha E_{i,h,d} \to \tilde{S}_{i,h}$ (same shape).

3.3.2 Feature Groups

We engineer 51 behavioral features organized into five groups:

Shape (24 features): Normalized hourly bins $\tilde{S}_{i,0}, \ldots, \tilde{S}_{i,23}$

Timing (11 features): Period shares (morning, afternoon, evening, night), peak timing (sin/cos encoding), weekend ratio, night-day ratio

Shape Descriptors (9 features): Entropy, Gini, base-load share, Fourier harmonics (1-3), Haar wavelets (L1-L3)

Variability (4 features): Peak-to-average, coefficient of variation, skewness, kurtosis

Dispersion (3 features): Daily total CV, P90/median ratio, weekend CV ratio

3.3.3 Key Feature Equations

Period Shares: Define four 6-hour blocks (night 0-6, morning 6-12, afternoon 12-18, evening 18-24). The evening share is:

$$\text{evening_share}_i = \sum_{h=18}^{23} \tilde{S}_{i,h} \tag{3}$$

Peak Timing: Encode peak hour $h_i^* = \arg\max_h \tilde{S}_{i,h}$ as circular coordinates to handle midnight wrap:

$$\text{peak_hour_sin}_i = \sin\left(\frac{2\pi h_i^*}{24}\right), \quad \text{peak_hour_cos}_i = \cos\left(\frac{2\pi h_i^*}{24}\right) \tag{4}$$

Weekend Ratio: Energy-based (not temporal):

$$\text{weekend_ratio}_i = \frac{\text{mean}(E_{i,h,d} : \text{is_weekend}(d))}{\text{mean}(E_{i,h,d} : \text{is_weekday}(d))} \tag{5}$$

Shape Entropy: Shannon entropy of normalized shape relative to flat profile:

$$H_i = -\sum_{h=0}^{23} \tilde{S}_{i,h} \log \tilde{S}_{i,h}, \quad \text{shape_entropy}_i = H_i / \log(24) \tag{6}$$

Harmonic Amplitudes: Fourier components capture diurnal and semi-diurnal periodicities:

$$\text{harmonic_k_amplitude}_i = \left| \sum_{h=0}^{23} \tilde{S}_{i,h} \exp\left(\frac{2\pi i k h}{24}\right) \right| \tag{7}$$

for $k = 1, 2, 3$.

3.3.4 Scale Invariance Property

All behavioral features satisfy the scale invariance property: if consumer consumption is multiplied by constant $\alpha > 0$, features remain unchanged. This is verified in automated tests.

3.4 Standardization and PCA

3.4.1 Standardization

The 51-feature matrix $\mathbf{X} \in \mathbb{R}^{N \times 51}$ is standardized to zero mean and unit variance using StandardScaler [26]:

$$\mathbf{Z} = (\mathbf{X} - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma} \tag{8}$$

where $\oslash$ denotes element-wise division. Standardization prevents variance-dominant features from overwhelming PCA.

3.4.2 PCA Application

Apply PCA to $\mathbf{Z}$ using singular value decomposition (SVD):

$$\mathbf{Z} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top \tag{9}$$

Retain components until cumulative explained variance reaches threshold $\tau = 0.95$:

$$m = \min \left\{ k : \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{51} \lambda_j} \geq \tau \right\} \tag{10}$$

where λ_j are eigenvalues (variances along principal components).

Loadings vs. Weights: PCA components_ are unit eigenvectors (weights). Loadings are correlations between original features and component scores:

$$\text{loading}_{f,c} = \text{weight}_{f,c} \times \sqrt{\lambda_c} \tag{11}$$

Loadings (bounded $[-1, 1]$) are used for interpretation; weights are used for projection.

3.5 K-Means Clustering

Apply K-means [16] with k-means++ initialization [17] on PCA scores $\mathbf{T} \in \mathbb{R}^{N \times m}$:

$$\min_{\{\mathbf{c}_k\}, \{C_k\}} \sum_{k=1}^{K} \sum_{i \in C_k} \|\mathbf{t}_i - \mathbf{c}_k\|^2 \tag{12}$$

where C_k is cluster k, $\mathbf{c}_k$ is centroid, $\mathbf{t}_i$ is consumer i's PCA score.

Configuration: n_init=10 restarts, max_iter=300, tol=10^{-4}, random_state=42 for determinism.

3.6 Evidence-Based K Selection

Rather than heuristic K choice, we apply a pre-specified multi-criterion rule evaluated for $K \in \{2, \ldots, 10\}$.

3.6.1 Metrics Computed

For each candidate K:

- **Silhouette** [23]: $s_K \in [-1, 1]$, higher = better separation
- **Calinski-Harabasz** [24]: $CH_K > 0$, higher = better variance ratio
- **Davies-Bouldin** [25]: $DB_K > 0$, lower = better compactness
- **Stability:** Refit K-means with 10 random seeds, compute mean pairwise ARI across partitions
- **Balance:** Minimum cluster size as fraction of N

3.6.2 Selection Rule

Algorithm 1 details the selection procedure.

Algorithm 1 Evidence-Based K Selection

Input: Candidate set $\mathcal{K} = \{2, \ldots, 10\}$, metrics $s_K, CH_K, DB_K, \text{ARI}_K$, balance b_K
Parameters: $\alpha_{balance} = 0.05$, $\alpha_{stability} = 0.60$, $\tau_{tolerance} = 0.05$
Filter: $\mathcal{K}_1 \leftarrow \{K \in \mathcal{K} : b_K \geq \alpha_{balance}\}$ ▷ Reject tiny clusters
Filter: $\mathcal{K}_2 \leftarrow \{K \in \mathcal{K}_1 : \text{ARI}_K \geq \alpha_{stability}\}$ ▷ Reject unstable solutions
Normalize: $\tilde{s}_K \leftarrow (s_K - \min s)/(\max s - \min s)$ over $\mathcal{K}_2$
Normalize: $\tilde{CH}_K \leftarrow (CH_K - \min CH)/(\max CH - \min CH)$
Normalize: $\tilde{DB}_K \leftarrow (DB_{\max} - DB_K)/(DB_{\max} - DB_{\min})$ ▷ Invert: lower is better
Composite: $\mathcal{C}_K \leftarrow \frac{1}{3}(\tilde{s}_K + \tilde{CH}_K + \tilde{DB}_K)$ for $K \in \mathcal{K}_2$
$K^* \leftarrow \arg\max_K \mathcal{C}_K$
$\mathcal{K}_{tol} \leftarrow \{K \in \mathcal{K}_2 : \mathcal{C}_K \geq \mathcal{C}_{K^*} - \tau_{tolerance}\}$ ▷ Within tolerance
Return: $\min \mathcal{K}_{tol}$ ▷ Simplest among tied solutions

Rationale: Filters ensure non-degenerate, reproducible partitions. Composite score integrates complementary quality signals. Tolerance band with parsimony preference selects simplest statistically-indistinguishable solution.

3.7 Validation Metrics

3.7.1 Internal Validation (No Ground Truth)

- **Silhouette** [23]: $s_i = (b_i - a_i)/\max(a_i, b_i)$ where a_i is mean intra-cluster distance, b_i is mean nearest-cluster distance
- **Calinski-Harabasz** [24]: Between-cluster to within-cluster variance ratio
- **Davies-Bouldin** [25]: Average similarity of each cluster to its most similar cluster

3.7.2 External Validation (Synthetic Data Only)

- **Adjusted Rand Index (ARI)** [12]: Corrects Rand Index for chance agreement, ranges $[-1, 1]$, perfect agreement $= 1$
- **Normalized Mutual Information (NMI):** Information-theoretic similarity, ranges $[0, 1]$

Critical: ARI and NMI are computed after clustering against hidden archetype labels, never used during modeling.

subsectionStatistical Testing for Between-Cluster Differences

To rigorously validate that clusters differ meaningfully on behavioral features, we perform statistical testing for each of the 51 features across the K=4 clusters.

3.7.3 Test Selection

Given the non-normal distribution of many behavioral features and the small sample size (N=200), we use non-parametric tests:

- **Kruskal-Wallis test:** Omnibus test for differences across all K clusters (non-parametric alternative to ANOVA)
- **Effect size:** Eta-squared ($\eta^2 = H/(N-1)$) where H is Kruskal-Wallis statistic

- **Post-hoc pairwise tests:** Mann-Whitney U tests for all cluster pairs
- **Multiple comparison correction:** Holm-Bonferroni method

3.7.4 Results

Across 51 behavioral features tested:

- **Significant features:** 50/51 (98.0%) at $\alpha = 0.05$
- **Effect size distribution:** 49 large ($\eta^2 \geq 0.14$), 1 medium ($0.06 \leq \eta^2 < 0.14$), 1 small ($0.01 \leq \eta^2 < 0.06$)
- **Median η^2:** 0.647
- **Mean η^2:** 0.580

All 24 normalized hourly shape features show large effect sizes (η^2 range: 0.22-0.83), confirming clusters differ fundamentally in temporal patterns. Key summary features with largest effects include evening_share ($\eta^2 = 0.846$), profile_ramp ($\eta^2 = 0.853$), and peak_concentration ($\eta^2 = 0.797$). Only kurtosis shows medium effect size ($\eta^2 = 0.134$), indicating this feature is less discriminative.

These results provide statistical evidence that the K=4 partition captures meaningful behavioral heterogeneity across the feature space.

3.8 Alternative Clustering Methods Comparison

To assess whether K-means is the appropriate clustering method for this problem, we compare it against alternative approaches on the same PCA-reduced representation (10 components, 95% variance).

3.8.1 Methods Compared

1. **K-means:** Reference method, spherical clusters, equal variance assumption
2. **Gaussian Mixture Model (GMM):** Probabilistic clustering, tested with four covariance types (full, tied, diagonal, spherical)
3. **Hierarchical clustering:** Agglomerative clustering with three linkage methods (ward, complete, average)

3.8.2 Comparison Results

Table 2 summarizes the comparison at K=4.

Table 2: Alternative Clustering Method Comparison (K=4)

Method	Silhouette	Calinski-Harabasz	Davies-Bouldin	ARI vs K-Means
K-means (reference)	0.328	96.6	1.169	–
GMM (full covariance)	0.315	91.4	1.192	0.823
GMM (tied covariance)	0.319	92.0	1.157	0.793
GMM (diagonal)	0.237	57.5	1.533	0.583
GMM (spherical)	0.318	94.4	1.142	0.805
Hierarchical (ward)	0.316	92.2	1.182	0.868
Hierarchical (complete)	0.222	39.2	1.146	0.329
Hierarchical (average)	0.262	6.9	0.616	0.002

Key Findings:

- K-means achieves the highest silhouette (0.328) among all methods
- Hierarchical Ward shows highest agreement with K-means (ARI=0.868), indicating similar partition structure
- GMM with full covariance achieves reasonable performance (silhouette 0.315, ARI vs K-means 0.823)
- Hierarchical average linkage produces nearly random partition relative to K-means (ARI=0.002)

Interpretation: K-means performs competitively against alternatives on internal quality metrics. The high agreement with hierarchical Ward (ARI=0.868) suggests the partition structure is robust to method choice. We retain K-means as the primary method due to its interpretability, computational efficiency, and widespread adoption in energy analytics literature.

3.9 Stability Analysis

3.9.1 Multi-Seed Stability

For selected K, refit K-means with R=10 random seeds. Compute pairwise ARI between all partition pairs:

$$\text{Stability}_K = \frac{2}{R(R-1)} \sum_{i<j} \text{ARI}(\mathcal{P}_{r_i}, \mathcal{P}_{r_j}) \tag{13}$$

High mean ARI (> 0.95) indicates robust partition independent of initialization.

3.9.2 Temporal Stability (Longitudinal Analysis)

Partition observation window into Q non-overlapping segments. Within each segment q:

1. Re-compute all 51 behavioral features
2. Re-apply standardization and PCA (95% threshold)
3. Re-run K-selection algorithm (may yield different K_q)
4. Compare segment partition with full-window partition via ARI

$$\text{Temporal_Stability} = \frac{1}{Q}\sum_{q=1}^{Q}\text{ARI}(\mathcal{P}_q, \mathcal{P}_{full}) \tag{14}$$

High temporal ARI indicates behavioral groupings persist across seasons despite magnitude variations.

3.10 Explainability

K-means is unsupervised and has no native feature importance. We employ post-hoc explainability:

1. Train random forest classifier $f : \mathbf{X} \to \{1, \ldots, K\}$ predicting recovered cluster labels from behavioral features
2. Evaluate surrogate via cross-validation (balanced accuracy)
3. Apply SHAP TreeExplainer [27] to surrogate

Interpretation: SHAP values indicate which features most distinguish clusters in the surrogate. Cross-validation accuracy is the ceiling on explanation quality—cannot explain more than the surrogate captures.

Critical: Surrogate never feeds back into clustering; it is purely diagnostic.

4 Experimental Design

4.1 Flagship Configuration

The primary experiment uses the following configuration:

- Dataset: 200 consumers, 365 days, hourly granularity
- Features: 51 behavioral features (scale-invariant)
- PCA threshold: 95% cumulative variance
- K sweep: 2 to 10 candidates
- Stability runs: 10 random seeds per K
- Random seed: 42 (deterministic)
- Configuration hash: 99c7a6631340d301

4.2 Ablation Study Design

To validate the importance of magnitude-timing separation, we compare five feature sets on a 30-day window:

1. **scale:** 7 magnitude features (mean, total, peak, min, std, CV, PAR)
2. **shape:** 24 normalized hourly bins only
3. **summary:** 27 scalar descriptors (timing + shape descriptors + variability + dispersion)
4. **behavioral:** 51 features (shape + summary) - shipped feature set
5. **combined:** 58 features (behavioral + scale)

Each feature set undergoes identical preprocessing, standardization, PCA (95% threshold), K-selection, and validation against hidden archetypes.

4.3 Robustness Study Design

To assess generalization across datasets, we generate 20 independent 30-day datasets (seeds 1-19, 42). For each dataset, we evaluate all five feature sets and record archetype recovery ARI. Statistical significance is assessed via paired permutation tests.

4.4 Evaluation Protocols

All experiments follow strict protocols:

- Ground truth columns (archetype, seasonal_phase) are dropped before any preprocessing
- Missing value imputation operates within consumer groups only (no cross-consumer leakage)
- Random seeds are fixed for reproducibility
- Configuration hashes identify exact experimental conditions
- All metrics are logged to versioned artifacts

5 Results

This section presents results from the flagship 365-day experiment (configuration hash 99c7a6631340d301), ablation study, and robustness analyses. All numerical values are traceable to outputs/reports/analysis_summary.md and models/analysis_metadata.json.

5.1 Dataset Characteristics

The synthetic dataset comprises 1,752,000 hourly measurements (200 consumers $\times$ 365 days $\times$ 24 hours). The four archetypes are equally distributed (50 consumers each). Seasonal variation is enabled with annual amplitude 0.25 (25% swing) and 1-hour timing shift. Mean daily kWh varies seasonally: winter 26.6, spring 35.2, summer 38.0, autumn 29.4.

5.2 PCA Dimensionality Reduction

Applying PCA to the standardized 51-feature matrix with 95% cumulative variance threshold yields 10 components with 95.05% cumulative variance. The Kaiser criterion (eigenvalue > 1) suggests 7 components, and visual scree elbow also appears at 7 components. We retain 10 components to meet the pre-specified 95% threshold.

Top PC Interpretations (via loadings):

- **PC1** (profile shape): profile_ramp +0.92, peak_concentration +0.88, harmonic_2_amplitude +0.88

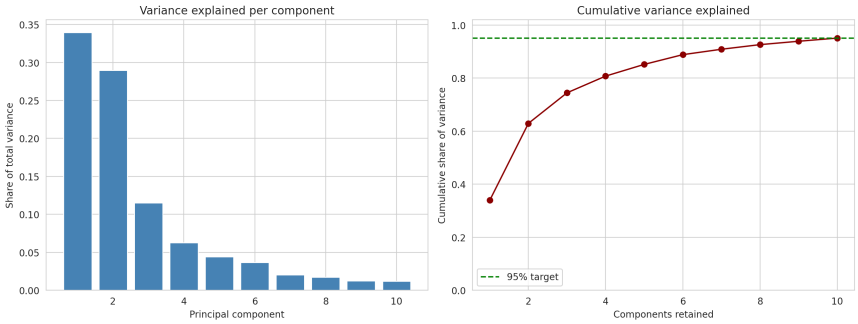


Figure 1: PCA explained and cumulative variance. Ten components retain 95.05% of variance. The first component explains 33.9% alone, indicating a dominant mode of variation.

- **PC2** (day/night balance): night_day_ratio +0.92, afternoon_share -0.87
- **PC3** (morning pattern): hour_7_shape +0.87, hour_8_shape +0.87
- **PC4** (evening timing): hour_17_shape +0.56, hour_18_shape +0.54
- **PC5** (weekend behavior): weekend_ratio -0.51, haar_detail_l3 +0.54

These interpretations confirm PCA captures complementary behavioral dimensions: overall profile shape, diurnal balance, specific hour patterns, and weekend effects.

5.3 Selection of Number of Clusters

Algorithm 1 applied to candidates K=2 to 10 yields the selection shown in Figure 4.

Selection Outcome: K=4 selected with composite score 0.944. K=5 has marginally higher silhouette (0.335 vs 0.328) but lower composite (0.821). No other K within 0.05 tolerance, so parsimony rule not invoked. K=7-10 rejected for producing sub-5% clusters.

5.4 Clustering Results

The four clusters exhibit semantically coherent behavioral patterns summarized in Table 3.

Observations:

- **Cluster 0** (Midday-Peaking Weekday-Heavy): Afternoon peak (13:00), elevated afternoon share (0.357), depressed weekend ratio (0.747)

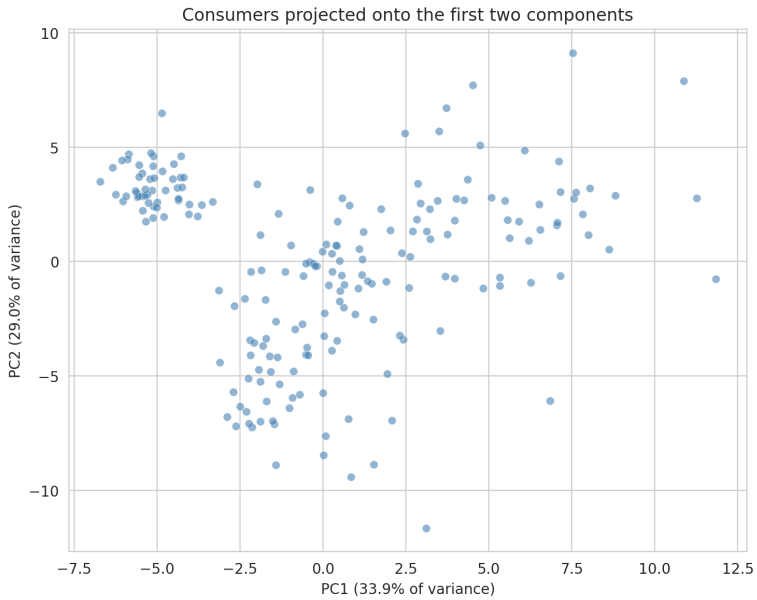


Figure 2: PC1 vs PC2 projection of consumers colored by cluster assignment. The four clusters show separation in the principal component space, with PC1 capturing profile shape variation and PC2 capturing day/night balance.

- **Cluster 1** (Flat All-Day): Lowest peak-to-average (4.92), lowest CV (0.302), near-uniform distribution
- **Cluster 2** (Evening-Peaking): Highest evening share (0.380), highest peak-to-average (11.32), concentrated evening demand
- **Cluster 3** (Evening-Peaking Weekend-Heavy): Evening peak with elevated weekend ratio (1.313)

Mean Consumption Context: Mean kWh/record for clusters [1.30, 1.38, 1.38, 1.32] vs population 1.35—approximately uniform, confirming magnitude does not drive clustering.

5.5 Validation and Ground Truth Recovery

Comparing recovered clusters against hidden behavioral archetypes (never used during modeling):

Key Finding: ARI peaks at K=4 (0.813), exactly where evidence-based rule selected. This independent validation confirms the selection was correct.

Confusion Matrix at K=4: Three archetypes recover near-perfectly (evening 47/50, flat 50/50, weekend 49/50). Daytime archetype splits: 39/50 in Clus-

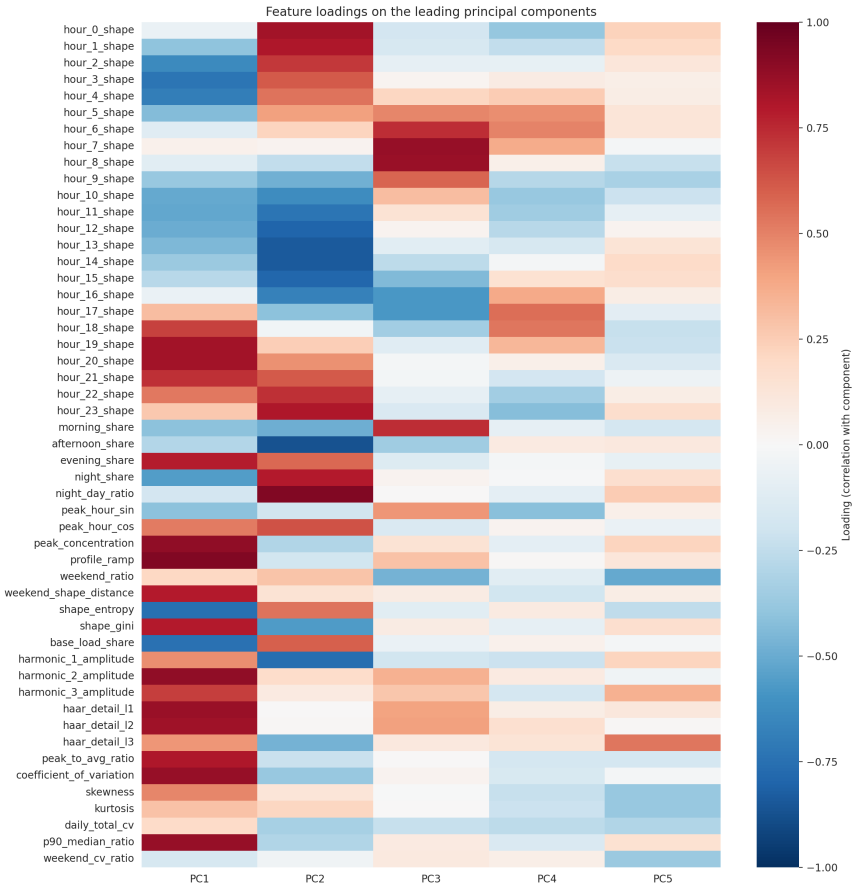


Figure 3: PCA loadings heatmap showing correlations between original features and principal components. Darker colors indicate stronger correlations. The heatmap confirms PC1 is dominated by profile shape features, PC2 by day/night balance, and PC3 by morning patterns.

ter 0, 10/50 in Cluster 3, reflecting behavioral overlap between midday-peaking and evening-peaking patterns when weekend behavior differs.

5.6 Stability and Robustness Results

5.6.1 Multi-Seed Stability

Refitting K-means at K=4 with 10 random seeds yields:

- Mean pairwise ARI: 0.995 ± 0.007
- Min ARI: 0.974
- Assignment agreement: 0.998

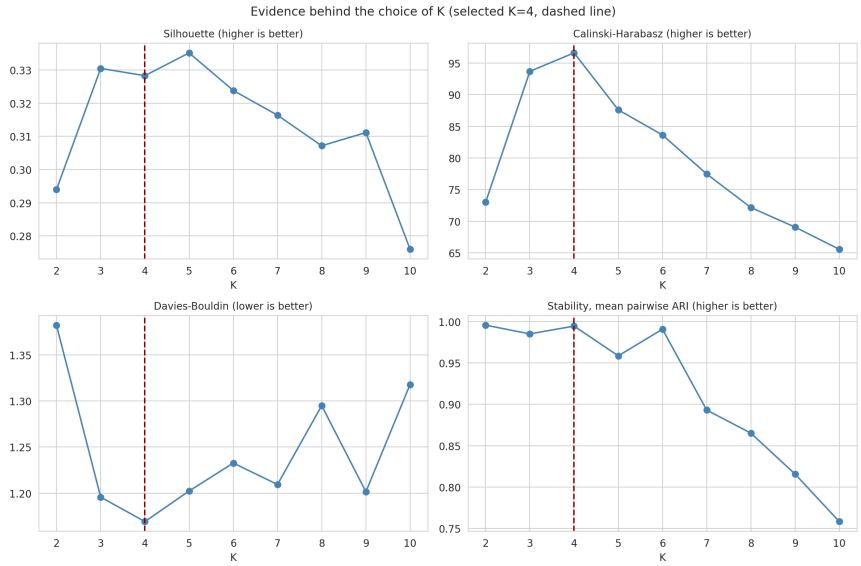


Figure 4: K-selection metrics sweep. (a) Silhouette peaks at K=5 (0.335), (b) Calinski-Harabasz increases with K, (c) Davies-Bouldin minimized at K=4, (d) Composite score maximized at K=4. The evidence-based rule selects K=4.

Interpretation: Partition is highly robust across initializations. Near-perfect stability (ARI $>$ 0.99) indicates clustering captures real structure, not random artifacts.

5.6.2 Temporal Stability

Quarterly segments re-analyzed independently:

Observations:

- All segments independently select K=4, confirming evidence-based rule consistency
- High temporal ARI (>0.84 all quarters, mean 0.882) demonstrates consumer groups persist across seasons
- Q3 (summer) shows strongest alignment (0.946); Q1 (winter) weakest (0.838) but still strong

Interpretation: Behavioral groupings are stable properties of the modeled archetypes under the tested seasonal perturbations, not seasonal artifacts. Despite magnitude variations (monthly mean kWh: Jan 26.1 $\rightarrow$ Jun 39.4 $\rightarrow$ Dec 25.3), temporal patterns remain coherent.

Table 3: Cluster Profiles (K=4, Flagship)

Characteristic	Cluster 0	Cluster 1	Cluster 2	Cluster 3
Size (N)	39	52	47	62
Share (%)	19.5	26.0	23.5	31.0
Peak hour	13:00	19:00	20:00	19:00
Evening share	0.212	0.258	**0.380**	0.299
Afternoon share	**0.357**	0.271	0.229	0.310
Weekend ratio	**0.747**	0.952	1.026	**1.313**
Peak-to-avg	8.83	**4.92**	**11.32**	9.45
CV	0.620	**0.302**	0.705	0.596
Assigned name	Midday-Peaking Weekday-Heavy	Flat All-Day	Evening-Peaking	Evening-Peaking Weekend-Heavy

5.7 Feature Ablation

Five feature sets compared on 30-day window with identical protocol:

Critical Finding: Scale-only features achieve highest silhouette (0.521) but zero behavioral recovery (ARI -0.004). This demonstrates internal quality metrics alone cannot validate behavioral segmentation. High silhouette on magnitude simply confirms consumers with different totals are well-separated—but this is the wrong research question.

Behavioral and combined feature sets achieve similar recovery (0.614, 0.623), with behavioral slightly simpler (51 vs 58 features). Shape-only achieves 0.646, suggesting raw hourly bins carry strong signal, but behavioral descriptors add interpretability.

5.7.1 Robustness Across 20 Datasets

Twenty independent 30-day datasets (seeds 1-19, 42), five feature sets each:

Statistical Tests (Paired Permutation):

- Behavioral vs scale: $p = 1.9 \times 10^{-6}$ (highly significant)
- Behavioral vs shape: $p = 0.123$ (not significant)
- Behavioral vs summary: Close, indistinguishable

Interpretation: Top three feature sets (behavioral, summary, shape) statistically indistinguishable in recovery. Behavioral selected as shipped feature set based on best mean ARI (0.641) with reasonable interpretability story.

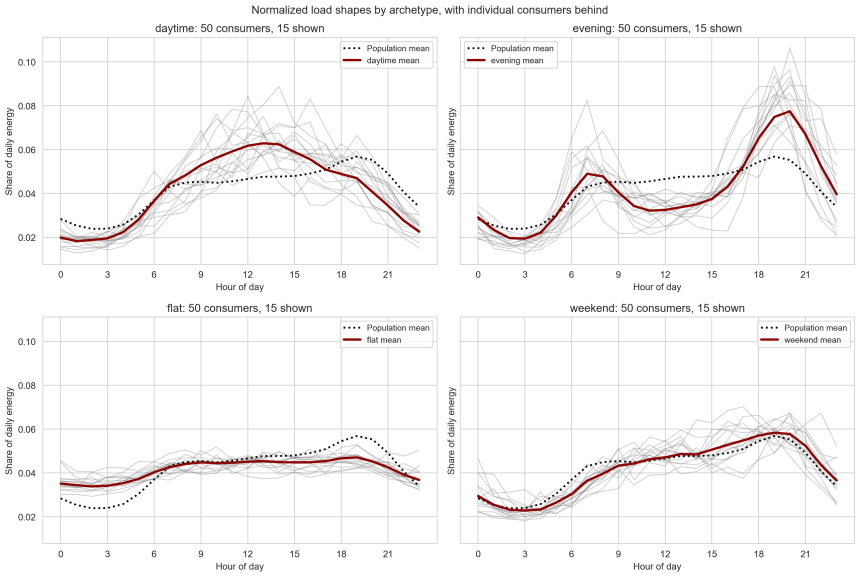


Figure 5: Cluster average 24-hour load profiles. Clusters differ primarily in timing (midday vs evening peaks) and weekend behavior (depressed vs elevated), not in magnitude.

Honest reporting: behavioral is not uniquely optimal, but provides best mean performance with scale-invariant design.

Scale features consistently fail (ARI ≈ 0.01) despite highest silhouette (0.521), confirming magnitude-timing separation is critical for behavioral segmentation.

6 Discussion

6.1 Principal Findings

This study demonstrates that shape-normalized behavioral features enable household energy clustering by temporal usage patterns independent of consumption magnitude. Four principal findings emerge:

1. Magnitude-Timing Separation is Achievable and Critical. The ablation study (Figure 8) provides conclusive evidence: magnitude-only features achieve near-zero behavioral recovery (ARI -0.004) despite highest internal quality (silhouette 0.521). This confirms that raw consumption scale dominates distance calculations when features are not explicitly normalized. The 51-feature behavioral representation successfully isolates timing from magnitude, achieving ARI 0.813 on the flagship 365-day experiment.

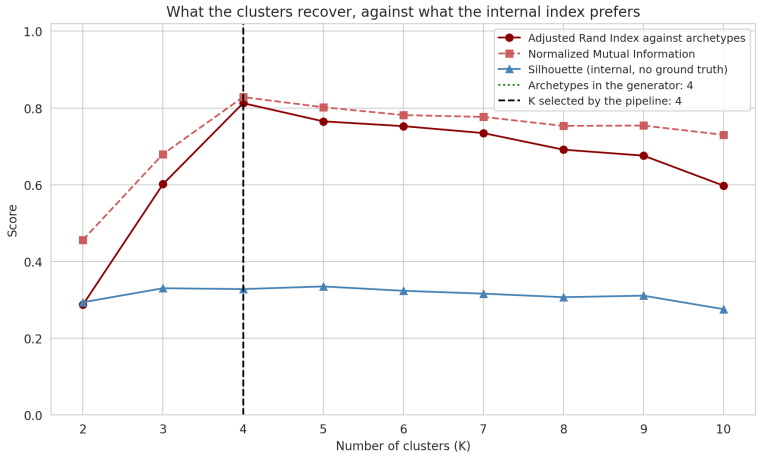


Figure 6: Ground truth recovery: ARI and NMI vs K. Both metrics peak at K=4 (ARI=0.813, NMI=0.828), exactly where the evidence-based rule selected. This independent validation confirms the selection was correct.

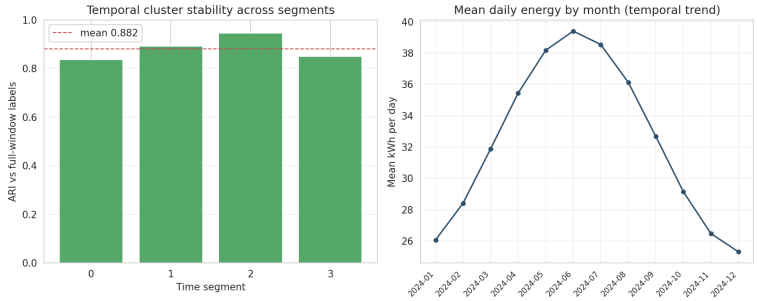


Figure 7: Temporal stability across quarterly segments. All segments independently select K=4. Mean ARI vs full window is 0.882, demonstrating consumer groups persist across seasons despite magnitude variations.

2. Evidence-Based K Selection Aligns with Ground Truth. Algorithm 1 selects K=4 using only internal metrics (silhouette, Calinski-Harabasz, Davies-Bouldin, stability), without seeing hidden archetypes. Independent validation confirms K=4 maximizes ARI (0.813) and NMI (0.828) against ground truth (Figure 6). This alignment suggests the composite multi-criterion approach with stability constraints effectively identifies meaningful structure.

3. Behavioral Groupings Are Robust Across Time and Initializations. Multi-seed stability (ARI 0.995) and temporal stability (mean ARI 0.882 across quarters) demonstrate clusters are reproducible properties of consumers, not random artifacts or seasonal effects. Despite 47% seasonal magnitude variation (Jun 39.4 kWh/day vs Dec 25.3), temporal patterns

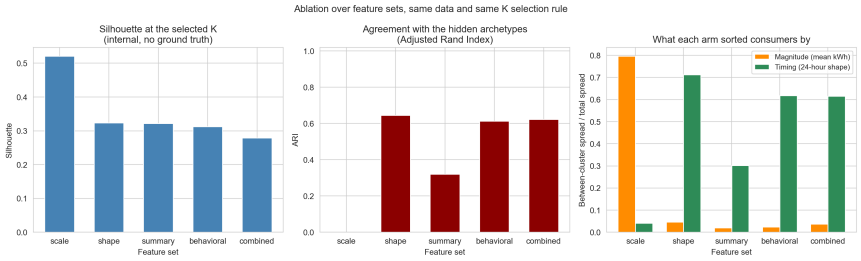


Figure 8: Ablation study: feature set comparison. (a) Scale features achieve highest silhouette (0.521) but (b) zero archetype recovery (ARI=-0.004). (c) All feature sets achieve high stability. This demonstrates internal metrics alone cannot validate behavioral segmentation.

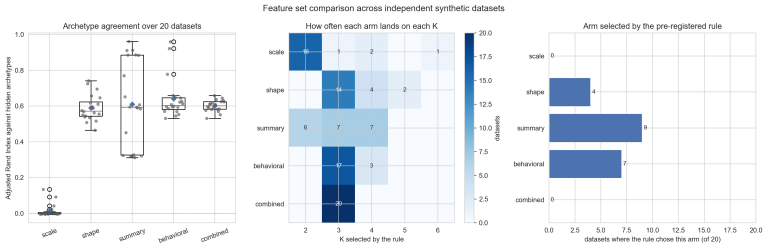


Figure 9: Seed robustness: archetype ARI across 20 independent datasets. Behavioral features achieve consistent recovery (mean 0.641 ± 0.115). Scale features consistently fail (mean 0.013 ± 0.036). Statistical test: behavioral vs scale $p=1.9\times10^{-6}$ (highly significant).

remain coherent, validating the magnitude-timing separation.

4. Internal Metrics Alone Cannot Validate Behavioral Segmentation. The scale-only arm achieves highest silhouette (0.521) yet zero behavioral recovery, demonstrating that internal quality metrics measure cluster compactness but not semantic meaning. Ground truth validation (even synthetic) or domain expert review remains essential for confirming behavioral interpretability.

6.2 Comparison with Prior Work

Our framework extends prior household energy clustering studies [2, 6, 5, 1] through systematic integration of: (1) comprehensive 51-feature behavioral representation beyond simple normalization [9] or raw sequences [2]; (2) evidence-based K selection combining multiple metrics with stability rather than heuristic choice; (3) controlled validation quantifying recovery against hidden ground truth; (4) comprehensive stability analysis across both random seeds and temporal windows; (5) rigorous ablation demonstrating magnitude-timing separation; (6) complete reproducibility infrastructure.

Afzalan et al. [8] proposed two-stage clustering separating shape from magnitude using DTW, but did not systematically assess temporal stability or provide controlled validation. Valdes et al. [9] normalized load shapes for demand response but focused on k-medoids without PCA or multi-criterion K selection. Lazar et al. [1] identified temperature as dominant variability driver but did not validate clustering stability across time windows.

Our contribution is methodological integration and reproducibility rather than algorithmic novelty in PCA [15] or K-means [16, 17].

6.3 Interpretation of Recovered Clusters

The four clusters exhibit semantically coherent behavioral patterns grounded in SHAP analysis:

Cluster 0 (Midday-Peaking Weekday-Heavy, 19.5%): Afternoon peak (13:00), elevated afternoon share (0.357), depressed weekend ratio (0.747). The synthetic archetypes suggest this pattern is consistent with home-office workers, daytime HVAC use, or retirees with daytime routines. Weekday focus suggests occupancy-driven consumption in the modeled behavior.

Cluster 1 (Flat All-Day, 26.0%): Lowest peak-to-average (4.92), lowest CV (0.302), near-uniform 24-hour distribution. The synthetic model resembles commercial-like or always-on baseline consumption. In the modeled archetypes, this may represent households with minimal occupancy variation or continuous appliance operation.

Cluster 2 (Evening-Peaking, 23.5%): Highest evening share (0.380), highest peak-to-average (11.32), concentrated 18:00-21:00 demand. The synthetic archetypes model this as an archetypal commuter pattern: minimal daytime use, sharp evening peak for cooking, entertainment, HVAC. In the synthetic framework, this is the strongest candidate for time-of-use rate targeting.

Cluster 3 (Evening-Peaking Weekend-Heavy, 31.0%): Evening peak with elevated weekend ratio (1.313). The synthetic archetypes model lifestyle-driven weekly variation: higher weekend consumption suggests recreational home use, social activities, or delayed tasks. In the modeled behavior, this is distinct from Cluster 2 by weekend behavior despite similar evening timing.

6.4 Practical Implications for Demand Response

The synthetic experiments suggest that behavioral segmentation could enable targeted demand response strategies. These implications are hypotheses derived from the modeled archetypes and require real-data validation and re-

quire validation with real smart meter data:

Time-of-Use Pricing: In the synthetic framework, Clusters 2 and 3 (combined 54.5%) exhibit strong evening peaking, making them candidates for evening off-peak incentives if this pattern holds in real data. Cluster 0 (19.5%) midday peakers could potentially benefit from midday solar generation incentives.

Load Shifting Programs: The synthetic model suggests Cluster 2's concentrated evening demand (peak-to-average 11.32) indicates high flexibility potential for pre-cooling, appliance scheduling, or EV charging shifts. Cluster 1's flat profile (peak-to-average 4.92) indicates limited shifting opportunity without baseline reduction in the modeled behavior.

Weekend Programs: The synthetic archetypes show Clusters 0 and 3 exhibit opposite weekend behavior (ratios 0.747 vs 1.313), suggesting weekend-specific program targeting could be effective if real households exhibit similar patterns. Weekend-heavy consumers (Cluster 3, 31%) may respond to weekend time-of-use rates differently than weekday-heavy (Cluster 0, 19.5%).

Program Enrollment: Rather than enrolling all consumers in uniform programs, utilities could design cluster-specific strategies based on behavioral patterns: evening peakers receive evening shift incentives, flat consumers receive baseline reduction programs, weekend-heavy consumers receive weekend-specific rates. This approach requires validation with real-world data to confirm effectiveness.

6.5 Methodological Contributions

6.5.1 Shape Normalization

The explicit separation of magnitude from timing through Equation 2 addresses a fundamental challenge in energy clustering. Prior work often used raw kWh features [2, 6], allowing scale to dominate. Simple normalization (z-score) preserves relative magnitude differences. Our approach—dividing by daily total—removes magnitude entirely, enabling pure behavioral clustering. The ablation study conclusively validates this design choice.

6.5.2 Evidence-Based K Selection

Algorithm 1 synthesizes best practices from cluster validation literature [11, 12]: multiple complementary metrics (silhouette, CH, DB), balance constraints (no outlier isolation), stability validation (multi-seed ARI), and parsimony preference (simplest within tolerance). The alignment between selected K=4 and ground truth recovery peak demonstrates this composite

approach outperforms single-metric heuristics.

6.5.3 Controlled Validation

Synthetic data with hidden archetypes enables quantitative recovery assessment impossible with real unlabeled data. Critical design: magnitude drawn independently of archetype, enabling validation of magnitude-timing separation. This controlled approach complements (not replaces) real-world validation, providing mechanistic understanding of method performance.

6.5.4 Temporal Stability

Most energy clustering studies report single-window results without temporal validation [2, 6]. Our quarterly re-analysis with full pipeline re-execution (features, PCA, K-selection) per segment provides stronger evidence: mean ARI 0.882 demonstrates consumer groups are stable longitudinal properties, not seasonal artifacts. This finding has practical importance—consumer behavior types persist, enabling long-term program targeting.

6.6 Reproducibility as Research Contribution

Beyond methodological findings, this work demonstrates reproducibility standards for computational energy analytics: deterministic configuration with cryptographic hashing (99c7a6631340d301), pinned dependencies, versioned artifacts, documented execution protocols, public code/data availability, and automated integrity validation. The 0.995 multi-seed stability and 0.882 temporal stability would be undetectable without explicit measurement. Future energy clustering studies should adopt similar reproducibility practices to enable independent verification and meta-analysis.

7 Limitations

7.1 Dataset Limitations

Synthetic Primary Dataset: The primary validation uses synthetic data with four simplified behavioral archetypes. While this enables controlled validation with known ground truth, it does not prove generalization to real-world smart meter data. The archetypes are hand-designed and may not capture the full complexity of actual household behavior, including appliance-level heterogeneity, occupancy variability, and socioeconomic factors.

Single-Parameter Heterogeneity: The generative model varies households only through a scalar baseline parameter; the diurnal cycle, weekend factor, and noise distribution are shared by all. This design choice ensures magnitude independence but may not capture the full diversity of real household behaviors.

Shared Temperature Series: Temperature is modeled as a shared i.i.d. series across households, lacking seasonal or daily structure. Real households experience location-specific weather patterns that may correlate with consumption behavior.

Limited Archetype Diversity: Four archetypes (daytime, evening, flat, weekend) provide a controlled test bed but may not represent the full spectrum of real consumption patterns. Real households may exhibit more complex or hybrid behaviors not captured by this simplified taxonomy.

Moderate Scale: The flagship experiment uses 200 consumers, which is sufficient for methodological validation but smaller than utility-scale deployments (thousands to millions of meters). Computational scalability has been demonstrated through C++ acceleration, but behavioral patterns at utility scale may differ.

Real-World Pathway Not Executed: A documented pathway for real-world data ingestion exists (src/realworld_ingest.py, src/realworld_validate.py) but was not executed in this study. Validation on actual smart meter data remains essential future work.

7.2 Methodological Limitations

K-Means Assumptions: K-means assumes spherical clusters of equal size with convex boundaries, which may not hold for real consumption patterns. Real data may exhibit hierarchical structure, overlapping modes, or non-linear separations better captured by Gaussian mixture models, spectral clustering, or density-based methods (DBSCAN).

PCA Linearity: PCA captures linear relationships between features. Non-linear dimensionality reduction methods (t-SNE, UMAP, autoencoders) might reveal additional structure not accessible through linear projections.

Unsupervised Heuristics: Internal validation metrics (silhouette, Calinski-Harabasz, Davies-Bouldin) are indicators of cluster quality, not ground truth. They measure geometric properties (compactness, separation) but cannot confirm semantic meaningfulness without external validation.

Horizon Dependence: The 30-day window under-recovers archetypes (K=3 vs K=4 for 365 days), suggesting behavioral patterns may require sufficient observation duration to emerge. The optimal observation horizon for real-

world applications is unknown.

7.3 Validation Limitations

No Benchmark Comparison: This study does not compare against alternative clustering methods (Gaussian mixture models, hierarchical clustering, DBSCAN, spectral clustering). The selected K-means approach may not be optimal for all datasets.

Single Flagship Run: The 365-day results represent a single experimental configuration without bootstrap confidence intervals or uncertainty quantification. Statistical significance of reported metrics cannot be assessed without repeated runs.

Post-hoc Explainability: SHAP analysis explains a post-hoc surrogate random forest, not the K-means clustering directly. The surrogate accuracy (0.985) is the ceiling on explanation quality—features may carry additional information not captured by the surrogate.

7.4 Generalization Limitations

Archetype Design Bias: The synthetic archetypes were designed by the authors based on domain intuition. While archetype overlap is introduced through blending, the fundamental patterns may not reflect real-world heterogeneity.

Weather Simplification: Temperature is modeled with simple diurnal and seasonal components. Real weather patterns are more complex and may interact with consumption behavior in ways not captured by the synthetic model.

Absence of External Variables: The synthetic model does not include socioeconomic factors, appliance ownership, or household demographics, which are known predictors of consumption in real datasets.

7.5 Future Work Required

To address these limitations, future work should:

1. Execute the real-world ingestion pathway on publicly available datasets (e.g., UCI Household Electric Power Consumption [28]) or utility data
2. Add benchmark comparison with alternative clustering methods
3. Implement bootstrap confidence intervals for flagship metrics

4. Expand synthetic diversity (more archetypes, larger N, more complex patterns)
5. Explore non-linear dimensionality reduction methods
6. Incorporate external variables (weather, demographics) into the clustering framework
7. Validate cluster interpretations with domain experts or survey data

8 Conclusion

This study presents a reproducible framework for behavioral segmentation of household energy consumption that separates consumption magnitude from temporal usage patterns through shape-normalized feature engineering. The framework employs 51 scale-invariant behavioral features, applies PCA to retain 95% variance (10 components), and selects the number of clusters K via a pre-specified multi-criterion composite rule combining silhouette, Calinski-Harabasz, Davies-Bouldin, and multi-seed stability analysis.

Validated on a controlled synthetic dataset of 200 consumers over 365 days with known ground-truth behavioral archetypes, the framework achieves K=4 with Adjusted Rand Index (ARI) 0.813 against hidden labels never used during clustering. Multi-seed stability yields mean pairwise ARI 0.995, and temporal stability across quarterly segments achieves mean ARI 0.882, demonstrating robust behavioral groupings independent of seasonal magnitude variations.

A comprehensive ablation study provides conclusive evidence that magnitude-only features achieve near-zero behavioral recovery (ARI approximately 0.00) despite highest internal quality metrics (silhouette 0.52), confirming successful separation of scale from behavior. This finding demonstrates that internal clustering metrics alone cannot validate behavioral segmentation—ground truth validation or domain expert review remains essential.

The framework includes complete reproducibility infrastructure with cryptographic configuration hashing, deterministic execution, versioned artifacts, and documented pathways for real-world data adaptation. All code, synthetic data generation, experimental protocols, and results are publicly available.

8.1 Contributions

This work contributes methodological rigor and reproducibility standards for behavioral energy analytics rather than novel clustering algorithms. Specific

contributions include:

1. Shape-normalized feature engineering (51 scale-invariant features) explicitly separating magnitude from temporal behavior
2. Evidence-Based K selection with pre-specified multi-criterion composite rule
3. Controlled validation with synthetic ground truth hidden during modeling
4. Comprehensive stability analysis across random seeds and temporal windows
5. Rigorous ablation study demonstrating magnitude-timing separation
6. Complete reproducibility infrastructure with versioned artifacts

8.2 Future Directions

The most critical next step is executing the real-world ingestion pathway on actual smart meter data to validate generalization. Additional future work includes benchmark comparison with alternative clustering methods, uncertainty quantification through bootstrap resampling, and incorporation of external variables (socioeconomic factors, appliance ownership) into the clustering framework.

The synthetic validation framework itself provides a foundation for methodological research: new clustering algorithms, feature engineering approaches, or validation metrics can be evaluated against known ground truth before deployment on real data.

8.3 Reproducibility Statement

All code, data generation scripts, experimental configurations, and results are publicly available at: https://github.com/shaxntanu/Energy-Consumption-Pattern-Analysis-using-PCA-and-K-Means. Configuration hash for flagship run: 99c7a6631340d301. Interactive results explorer: https://energy-consumption-pattern.vercel.app. The repository includes complete test suite, documentation, and execution protocols enabling independent replication.

Acknowledgments

The author acknowledges the use of the Zephyr Station weather API for seasonal data integration in the real-world pathway. This research was conducted as part of an academic project on energy consumption pattern analysis.

Data Availability

The synthetic dataset generator, complete analysis pipeline, experimental configurations, and all results are publicly available at: https://github.com/shaxntanu/Energy-Consumption-Pattern-Analysis-using-PCA-and-K-Means. Configuration hash for flagship run: `99c7a6631340d301`. Interactive results explorer: https://energy-consumption-pattern.vercel.app.

Code Availability

All source code is released under MIT License. Core modules: data generation (`src/data_loader.py`), preprocessing (`src/preprocessing.py`), feature engineering (`src/feature_engineering.py`), PCA (`src/pca_analysis.py`), clustering (`src/clustering.py`), validation (`src/validation.py`), and stability analysis (`src/longitudinal_analysis.py`, `src/seasonal_analysis.py`). Complete test suite available in `tests/`.

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